Semi-Supervised 3D Shape Reconstruction with DeepSDF

AI6131 · NTU MSAI · April 2026

"Reducing annotation requirements for implicit neural shape learning"

Research Question

> **Can we train a DeepSDF auto-decoder with fewer labelled SDF samples while maintaining reconstruction quality?**

Motivation

- Generating SDF labels requires **dense normal-offset sampling** — expensive per shape
- Unlabelled points (uniform sphere probes) need no annotation, yet constrain the field via $\mathcal{L}_{\text{eik}}$
- Reducing the labelled fraction enables scaling to larger shape collections

Research Variables

Variables under study

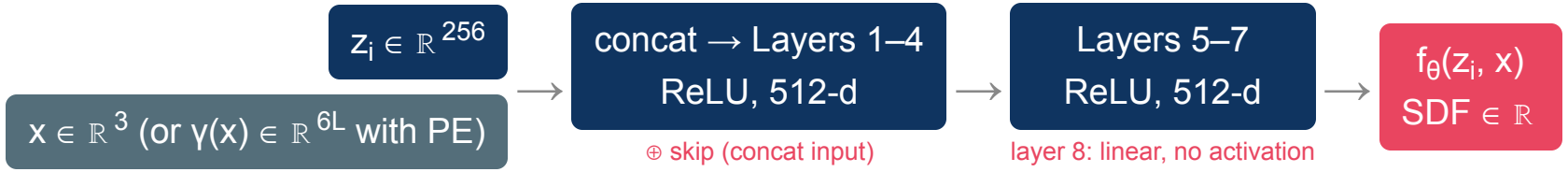
Variable	Values tested
Supervision ratio	100% → 50% → 10% → 5%
Eikonal regularization	On / Off
Positional encoding	On / Off (Fourier, details in experiment table)

Background: DeepSDF Auto-Decoder

Core idea

Each shape i gets a latent code $\mathbf{z}_i \in \mathbb{R}^{256}$.

A shared decoder $f_\theta(\mathbf{z}_i, \mathbf{x})$ maps **(latent, 3D point) → SDF value**.



- **8-layer MLP**, hidden dim 512, skip at layer 4
- No output activation — raw SDF value
- Last-layer sphere-biased init: small weight std + bias = 0.1; interior layers use Xavier

Training objective

$$\min_{\theta,\, \{\mathbf{z}_i\}} \sum_i \mathcal{L}_i(\theta, \mathbf{z}_i)$$

- **No encoder** — auto-decoder framework
- θ and all $\{\mathbf{z}_i\}$ optimized **jointly** with Adam
- This project: **train-set reconstruction only**
  (test-time latent opt. out of scope)

Semi-Supervised Extension

Semi-supervised objective

$$\mathcal{L} = \mathcal{L}_{\text{sdf}} + \lambda(t) \cdot \mathcal{L}_{\text{eik}}$$

($\lambda_z \cdot \mathcal{L}_z$ latent regularization applied in all runs — standard DeepSDF component)

Supervised points — contribute to L_sdf + L_eik

Surface samples **offset along normals** at ε ∈ {0.005, 0.01, 0.05},
multiplier $j \in \{-2, -1, +1, +2\}$.
SDF label: $s_j = j \times \varepsilon$ — **signed** approximate offset, not ray-cast GT.

$$\mathcal{L}_{\text{sdf}} = \frac{1}{N_s} \sum_{j \in S} \left| f_\theta(\mathbf{z}, \mathbf{x}_j) - s_j \right|$$

Unsupervised points — contribute to L_eik only

Sampled **uniformly in the unit sphere** (rejection sampling). No SDF label — constrain the field geometry via the Eikonal term.

$$\mathcal{L}_{\text{eik}} = \frac{1}{N} \sum_j \left(\|\nabla_{\mathbf{x}} f\|_2 - 1 \right)^2$$

Semi-Supervised Extension

Eikonal warmup — $\lambda(t) = \lambda_{\text{eik}} \cdot \min\left(1,\, t/t_{\text{warmup}}\right)$

Supervision	Warmup	Rationale
100% or 50%	100 ep	Enough supervised signal from start
10%	150 ep	Prevent L_eik dominating sparse labels
5%	200 ep	Maximum warmup for minimal supervision

Experiment Design

Dataset: ShapeNet — 300 shapes (airplane / chair / table), 75% used for training (train_split=0.75) · 250K unsupervised + up to 250K supervised points per shape (supervised count = 250K × ratio) · ~6h per run on TC2

Exp	Supervision	Eikonal	Positional Enc.	Seeds
EXP-01	100%	No	No	1
EXP-02	100%	Yes	No	1
EXP-03	50%	Yes	No	1
EXP-04	10%	Yes	No	3
EXP-05	5%	Yes	No	1
EXP-06	10%	Yes	Yes (L=6)	3

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Exp	Supervision	Eikonal	Positional Enc.	Seeds
EXP-07	5%	Yes	Yes (L=6)	1
EXP-08	10%	Yes	Yes (L=6) + $\mathcal{L}_{\mathrm{2nd}}$	1
EXP-09	100%	Yes	Yes (L=6)	1
EXP-10	100%	Yes	Yes (L=4)	1
EXP-11	10%	Yes	Yes (L=4)	1
EXP-12	5%	Yes	Yes (L=4)	1

Primary comparisons: EXP-01 → EXP-02 (Eikonal effect) · EXP-02 → EXP-04 → EXP-05 (label reduction) · EXP-04 vs EXP-06 (PE effect)

Preliminary Results

Preliminary — evaluation bugs discovered after these runs; see following slides. Non-PE group: original evaluator (MC res=128). PE group: rerun with fixed evaluator (MC res=128). Cross-group CD/NC comparison is not valid.

No positional encoding — original evaluator (MC res=128)

Exp	Supervision	CD mean	NC mean	Seeds
EXP-01	100%	0.0593	0.5522	1
EXP-02	100% + Eik	0.0543	0.5920	1
EXP-03	50% + Eik	0.0534	0.5924	1
EXP-04	10% + Eik	0.0496	0.5987	3
EXP-05	5% + Eik	0.0509	0.5766	1

Preliminary Results

With positional encoding — rerun with fixed evaluator (MC res=128)

Exp	Supervision	PE	CD mean	NC mean	Seeds
EXP-06	10% + Eik	L=6	0.1399	0.5080	3
EXP-07	5% + Eik	L=6	0.1448	0.5074	1
EXP-08	10% + Eik	L=6+$\mathcal{L}_{2\mathrm{nd}}$	0.1443	0.5053	1
EXP-09	100% + Eik	L=6	0.1450	0.5031	1
EXP-10	100% + Eik	L=4	0.1401	0.5077	1
EXP-11	10% + Eik	L=4	0.1427	0.5071	1
EXP-12	5% + Eik	L=4	0.1400	0.5073	1

Lower CD is better · Higher NC is better · Non-PE group: variable MC success (EXP-02: 261/300, EXP-03: 259/300, EXP-05: 295/300; EXP-04 seed42: 263/300) · EXP-04 per-seed CD means: {0.0609, 0.0443, 0.0436}; EXP-06 per-seed CD means: {0.1443, 0.1375, 0.1380}

Evaluation Pipeline Review

Before drawing conclusions from the results, I audited the evaluation pipeline. The main issue was incorrect shape→latent-index assignment: the evaluator used sorted alphabetical shape order, but training assigns latent indices in training-loop order, so each shape was decoded with the wrong latent code. Several secondary issues were also identified.

- Evaluation is now restricted to training shapes with the correct stored latent indices
- Secondary fixes applied: checkpoint selection criterion, train/val split ordering
- PE runs (EXP-06–12) have been re-evaluated with the corrected evaluator; non-PE results use the original

Key Evaluation Issue: Latent Code Assignment

> DeepSDF is an auto-decoder: latent codes are learned one per training shape, indexed by training order. The evaluator must use the same shape→latent-index mapping that training used. If it uses a different ordering, it retrieves the wrong latent for each shape and the resulting CD/NC values do not reflect reconstruction quality.

Before

- Evaluator iterated shapes in sorted alphabetical order
- Latent index assigned by sorted position, not training position
- Wrong latent retrieved for each shape → CD/NC reflects ordering mismatch, not shape quality

After

- Evaluator loads `train_shapes.json` to reconstruct exact training-order index assignment
- Val shapes have no stored latent; TTO used if val split is evaluated
- Reported CD/NC reflects actual reconstruction of training shapes

Reported metrics are train-set reconstruction quality — held-out generalization is not evaluated here

Current Status and Next Step

Established

- The 12-experiment design remains unchanged
- Checkpoints for the planned experiment matrix already exist
- The quantitative evaluation has been corrected for re-validation

Not yet claimable

- Quantitative comparisons across experiments are still being re-validated
- No comparative conclusion is claimed until re-evaluation is complete

Next step: Re-validate the existing checkpoints under the corrected evaluation, then report the quantitative comparison across experiments.